

ENGINEERING PORTFOLIO

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Embedded systems, computer vision, and physical prototypes

Chess move tracking camera

SPIS 2026 · Working team project

Visual changes + known board state

Calibrated square geometry links image locations to the board.
The previous game state and chess constraints narrow candidate moves.

Testing under disturbances

We tested shadows, camera shaking, bumped pieces, and imperfect placement.

Next step

Preserve labeled trials and source code so the result can be reproduced.

Tracked line-following robot

PLTW 2024 · Team hardware prototype

Hardware

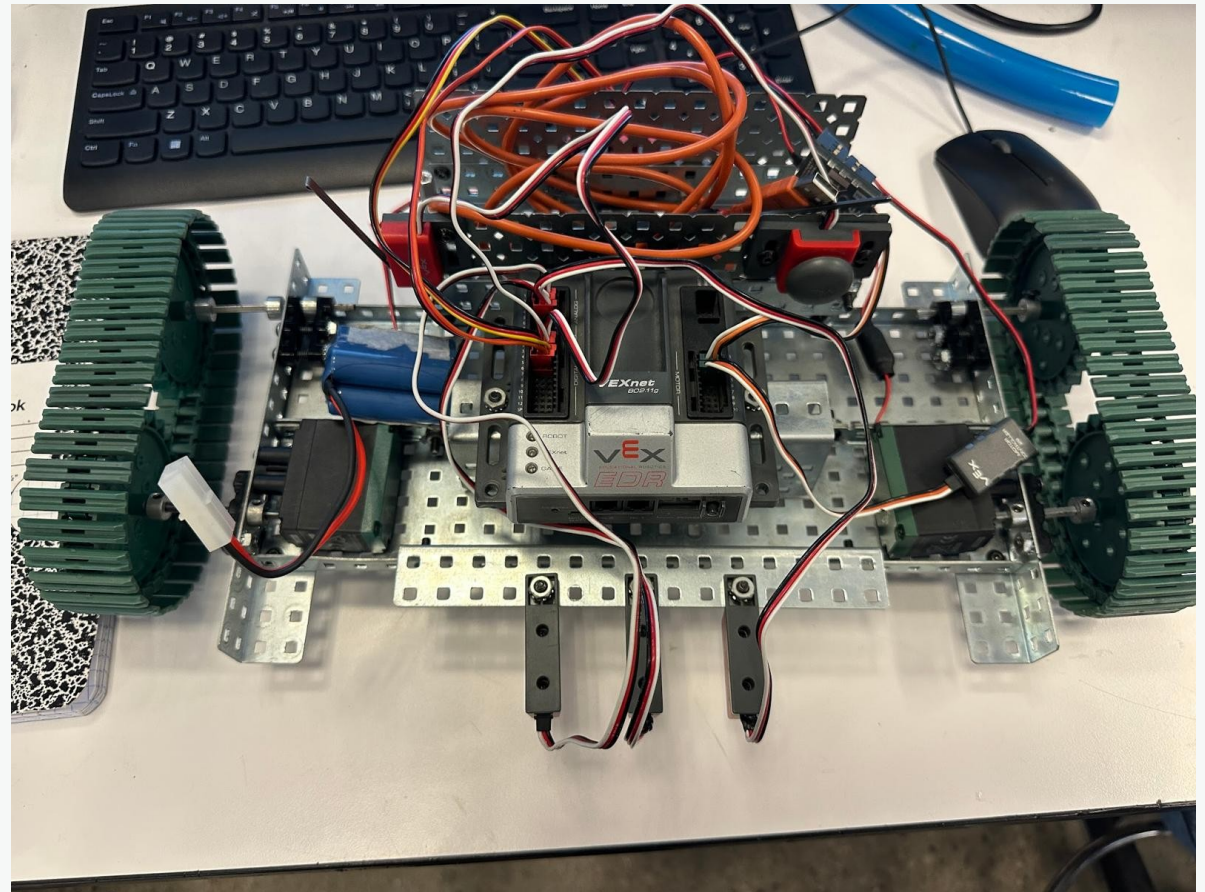
VEX Cortex, two motors, tracks, three line trackers, and ultrasonic sensing.

Integration

Built the chassis and integrated sensors, switching, and wiring. RobotC control work remained incomplete.

Lesson

Start software trials before the mechanical build is finished.



Environment-adaptive speaker

ECE Makerspace 2026 · Individual design proposal

Enclosure concept

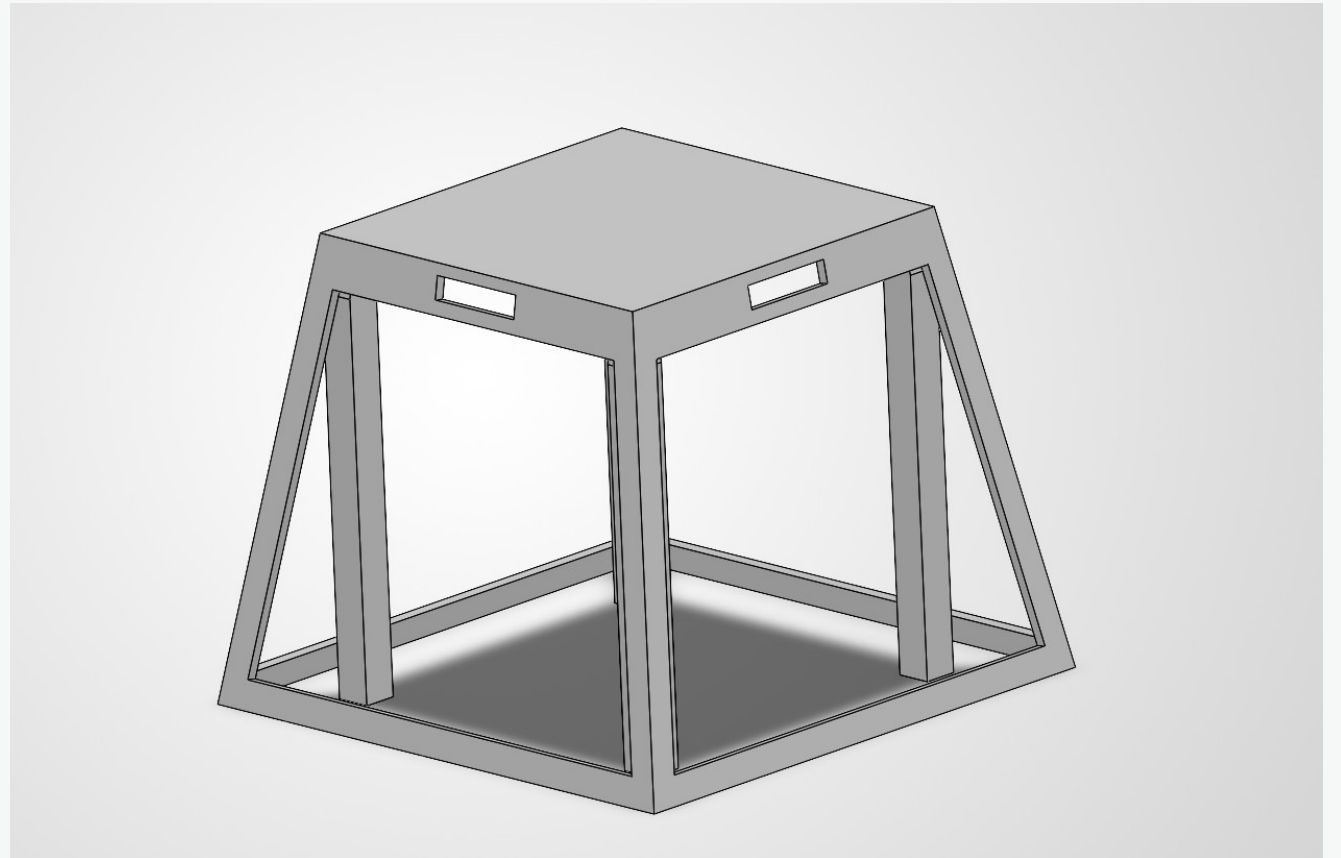
A 6×6 in base and 4.5 in height. Four posts support the truncated pyramid frame.

Adaptive behavior

Each face would sense nearby obstructions and reduce its corresponding driver output.

Completed work

Frame CAD, bottom fastener locations, and a construction plan.



Speaker architecture and testing

Four independent outputs remain a design requirement

Audio control

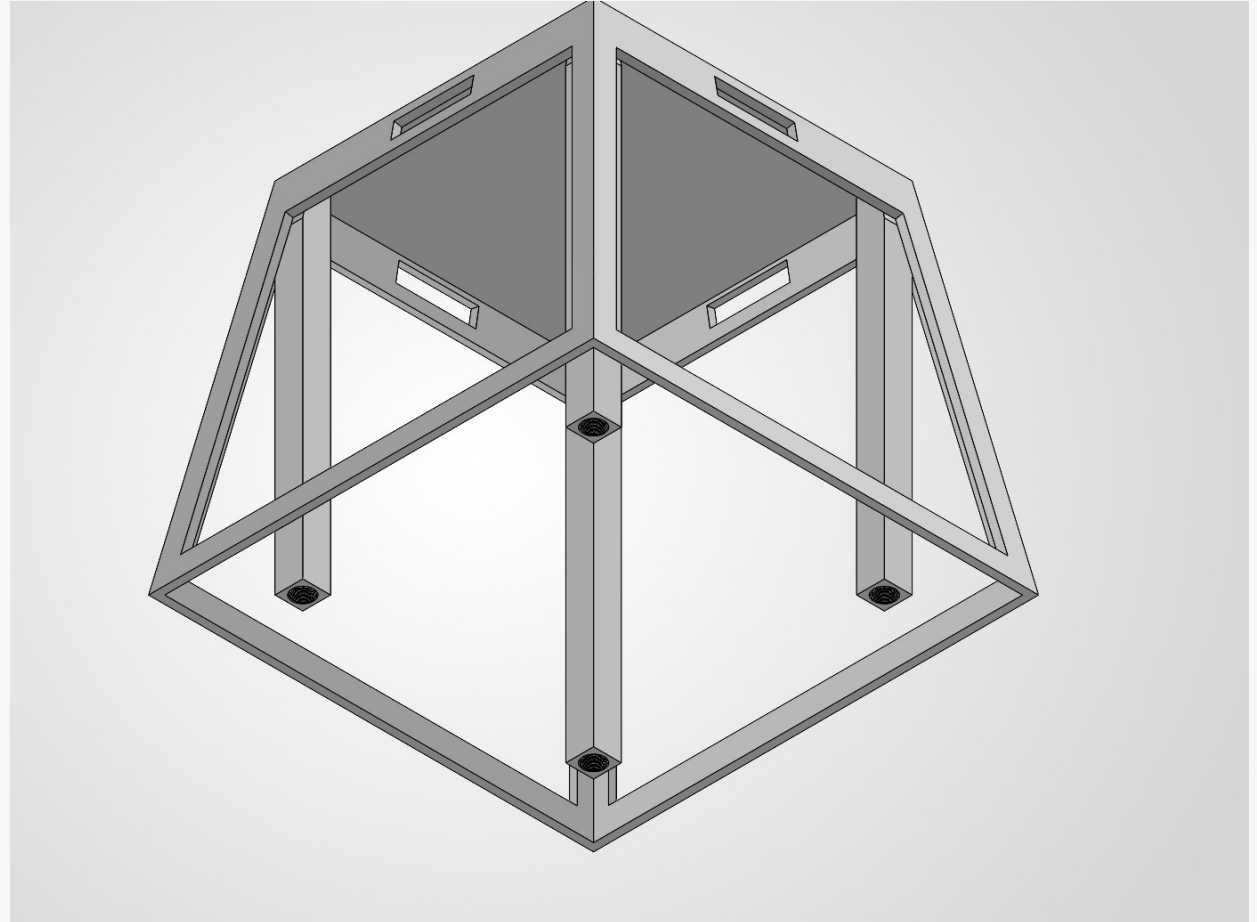
The listed stereo amplifier needs revision or added channel control for four independent outputs.

Test plan

Compare adaptive mode on and off near walls and corners. Measure power and audible effects.

Design questions

Sensor thresholds, driver interaction, and internal volume still need testing.



Minecraft chess set

PLTW 2024 · Individual parts in a team assembly

My parts

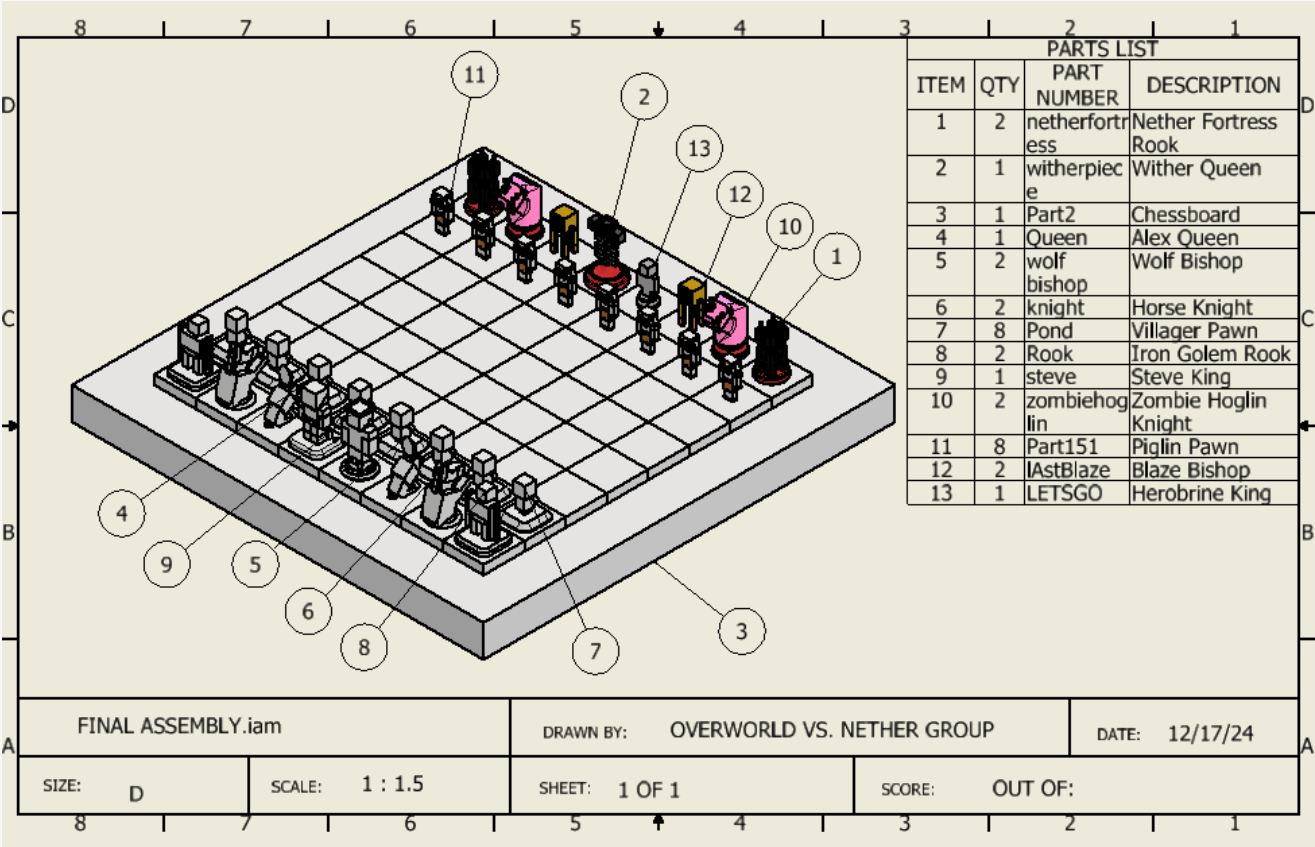
Knight, Steve piece, and wolf bishop. Drawing title blocks identify my contribution.

Documentation

Dimensioned views and neutral STEP geometry preserve the designs.

Next iteration

Check manufacturability and base consistency with a physical test print.



Star container and CAM studies

PLTW 2025 · CAD models and toolpath simulations

Part geometry

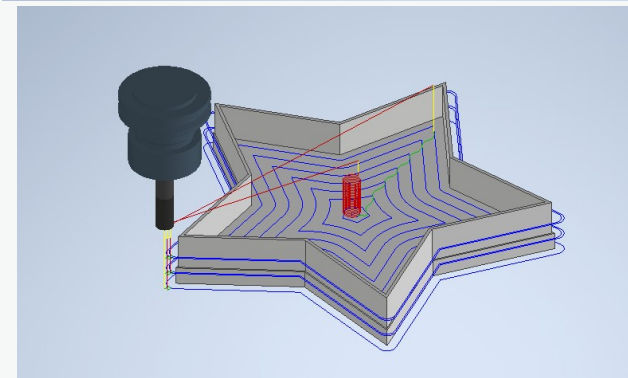
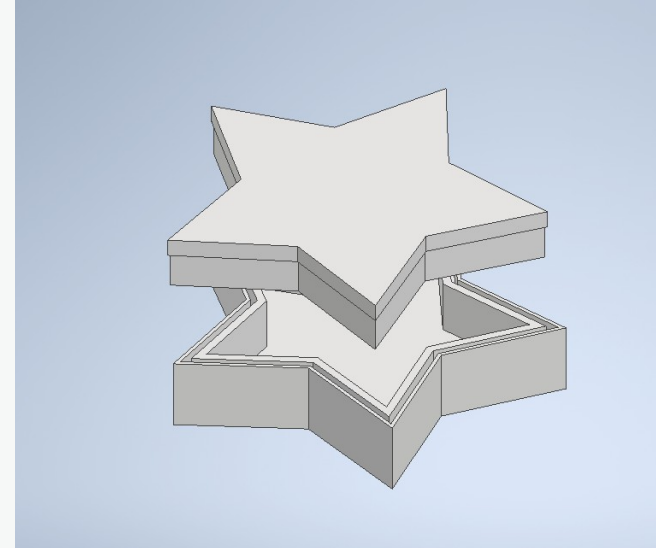
A star-shaped base and fitted lid.
Sketches examine the fit between
the opening and the lid.

CAM work

2D pockets and contours, with
related studies in facing, slotting,
drilling, and surface machining.

Lesson

Plan tool access while modeling.
Reduce unnecessary tool travel.



Insole design study

Senior PLTW capstone · Team planning and design research

Material stack and prototype planning

Compared foam, silicone, fabric, and memory foam.
Planned cutting, shaping, and assembly with available equipment.

Sensing exploration

Considered load cells, HX711 modules, and a microcontroller. Sensor integration remained a possible extension.

Next iteration

Evaluate the material prototype, then calibrate one sensor before designing a wearable layout.

Bridge build and design reflection

PLTW 2024 · Individual physical build

Constraints

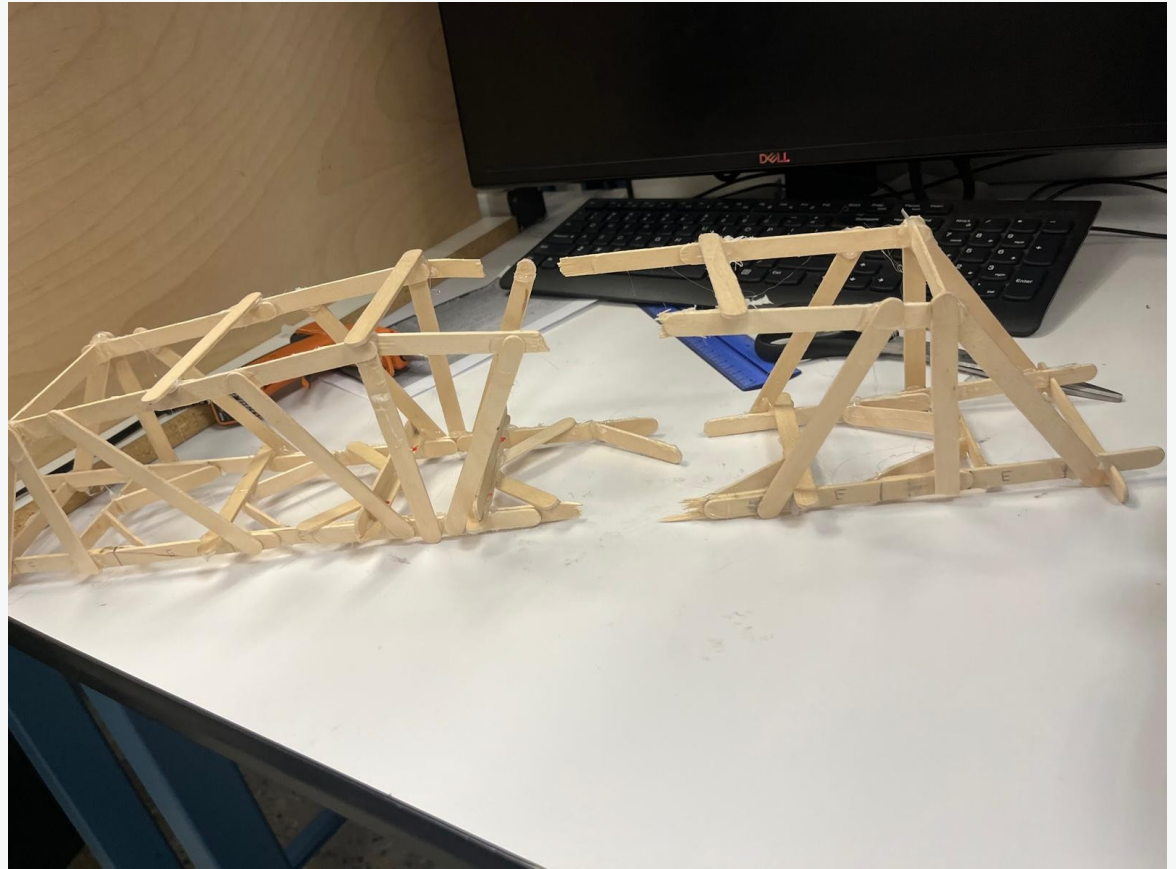
At most 60 popsicle sticks and a required 24-inch span.

Outcome

The bridge carried a load but fell about four inches short of the required span.

Lesson

Verify dimensions before bracing. A stronger structure still has to meet its original requirements.



What I want to build next

Embedded systems that sense the physical world and respond reliably.

I'm developing my C and embedded programming skills.
I'm looking for robotics, hardware, and research teams where I can contribute while continuing to learn.

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github.com/pur35t/Brief-Projects-